



PT.SUMITOMO ELECTRIC WINTEC INDONESIA

No. Dokumen	4-QAS-PP07-001
Revisi	00
Tgl Berlaku	2-Dec-16

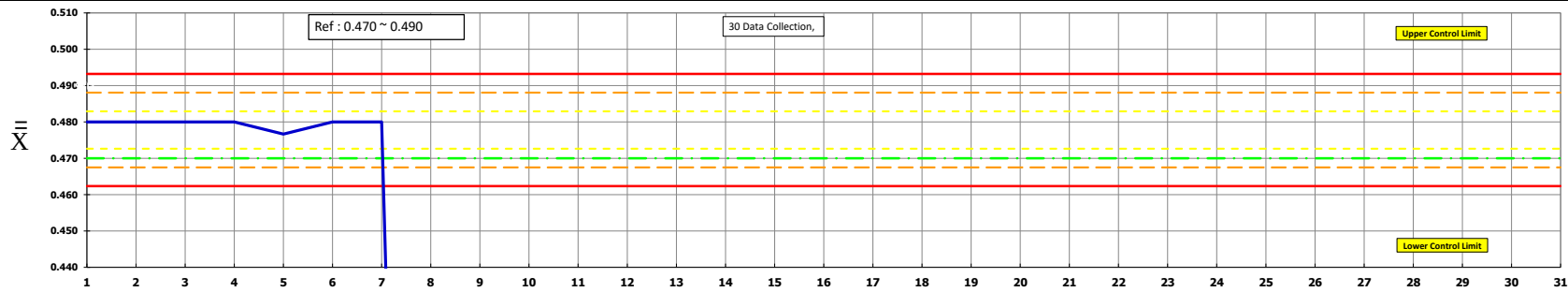
 $\bar{X}$ - R CONTROL CHART

MACHINE Softability	SECTION QC	OPERATION Softability	DATE CONTROL LIMITS CALC.	SPECIFICATION min max 0.460	COURSE NO.	CONTROL ITEM (Δ) YES ☐ NO ☐
MONTH May-25	CHARACTERISTIC Kelenturan	SAMPLE SIZE / FREQUENCY n = 3	TYPE / SIZE : D2DSW2 / 0.50	PIC : YAYAT S.		

$$\bar{X} = \text{AVERAGE } \bar{X} = 0.470$$

$$U.C.L. = \bar{X} + A_2 \bar{R} = 0.4932$$

$$L.C.L. = \bar{X} - A_2 \bar{R} = 0.4623$$

AVERAGE ($\bar{X}$ BAR CHART)

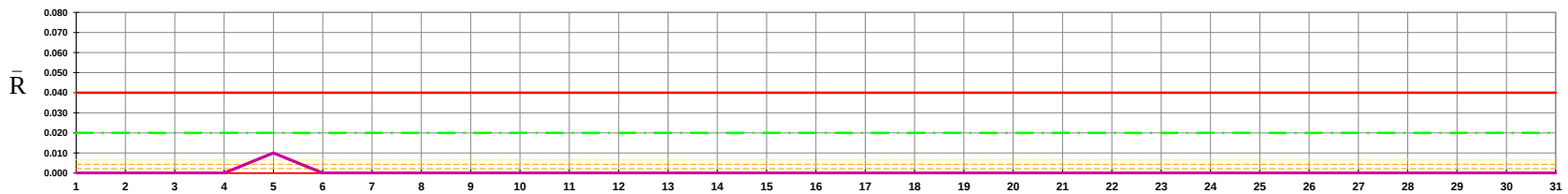
$$\bar{R} = \text{AVERAGE } R = 0.020$$

$$U.C.L. = D_4 \bar{R} = 0.040$$

$$L.C.L. = D_3 \bar{R} = 0.00$$

RANGES ($\bar{R}$ - CHART)

Special Cause Marker Legend:

Test 1 ● Test 2 ◆ Test 3 ▲ Test 4 ■

SPECIAL CAUSES
☒ TEST 1: ANY POINT OUT OF CONTROL LIMIT
☒ TEST 2: 7 POINTS IN A ROW ON SAME SIDE OF CENTER LINE
☒ TEST 3: 6 POINTS IN A ROW ALL INCREASING OR DECREASING
☒ TEST 4: 14 POINTS IN A ROW, ALTERNATING UP & DOWN
☒ TEST 5: 2 OUT OF 3 POINTS > 2 STDEV FROM CENTER LINE (SAME SIDE)
☒ TEST 6: 4 OUT OF 5 POINTS > 1 STDEV FROM CENTER LINE (SAME SIDE)
☒ TEST 7: 15 POINTS IN A ROW WITHIN 1 STDEV OF CENTER LINE (EITHER SIDE)
☒ TEST 8: 8 POINTS IN A ROW > 1 STDEV FROM CENTER LINE (EITHER SIDE)

ACTION INSTRUCTION

- _____
- _____
- _____
- _____
- _____

SUBGROUP SIZE	A ₂	d ₂	D ₃	D ₄
2	1.881	1.128	0	3.267
3	1.023	1.693	0	2.574
4	0.729	2.059	0	2.281
5	0.577	2.326	0	2.115
6	0.483	2.534	0	2.004
7	0.4193	2.704	0.0756	1.9244
8	0.3726	2.847	0.1361	1.8639

Special Cause	Titik berada di luar garis merah atau lebih dari 3 σ		6 titik berada dalam satu urutan secara terus menerus, baik posisi naik ataupun turun		4 atau 5 titik lebih dari 1 σ dengan posisi yang sama dan titik keluar dari garis merah atau 3 σ	
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Month																																
DATA	Date	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
	1	0.48	0.48	0.48	0.48	0.48	0.48	0.48																								
	2	0.48	0.48	0.48	0.48	0.48	0.48	0.48																								
	3	0.48	0.48	0.48	0.48	0.47	0.48	0.48																								
N	3	3	3	3	3	3	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
AVERAGE, X	0.48	0.48	0.48	0.48	0.48	0.48	0.48	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	
Range	0.000	0.000	0.000	0.000	0.010	0.000	0.000																									
Moving Range		0.002	0.768	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	
SHIFT 1	yat	yat	yat		yat		yat																									
SHIFT 2																																

CALCULATION

$$\bar{X} = 0.4795$$
$$\bar{R} = 0.0100$$
$$S_{\bar{X}} = 0.0059 \quad S_{\bar{R}} = 0.002$$

THE PROCESS MUST BE IN CONTROL
BEFORE CAPABILITY CAN BE DETERMINED

$$C_{pk} = \min \left\{ \frac{\bar{X} - LSL}{3S_{\bar{X}}}, \frac{USL - \bar{X}}{3S_{\bar{X}}} \right\}$$
$$= \min \{ 0.97 \quad 0.77 \}$$
$$= 0.77$$

$$C_p = \frac{USL - LSL}{6S_{\bar{X}}}$$

$$= 0.87$$

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